C++

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Switch…case statement –

```
switch(expression){
   case constant-expression  :
      statement(s);
      break; /* optional */
   case constant-expression  :
      statement(s);
      break; /* optional */
      /* you can have any number of case statements */
   default : /* Optional */
      statement(s);
}
```

The following rules apply to a switch statement:

- Works only on integer, character and enum values.
- Data type of case value and the variable in the switch must match.
- When a break statement is reached, the switch terminates, the control jumps outside switch structure.
- More readable and structured as compared to if..else if..else ladder.
- A switch within switch is possible, which is called nested switch statements.
- Very useful while writing menu driven programs.

```
#include<iostream>
using namespace std;
int main()
{
	int choice;
	cout<<"Enter your choice : ";
	cin>>choice;
	switch(choice)
	{
		case 1:
			cout<<"First\n";
		case 2:
			cout<<"Second\n";
		case 3:
			cout<<"Third\n";
		default:
			cout<<"Wrong choice\n";
	}
	return 0;
}
```

Output:

```
Enter your choice : 1
First
Second
Third
Wrong choice
```

Sample code to understand switch..case

```
switch(ch)
	{
		case 'a':
		case 'e':
		case 'i':
		case 'o':
		case 'u':
			cout<<"Alphabet is a vowel\n";
			break;
		default:
			cout<<"Alphabet is a consonant\n";
	}
```

Loop Statement

When set of statements to be executed multiple times, loop is used. It is also called as iterative statement. There are 3 types of loops supported by C Compiler

1. for(..;..;..) loop
2. while(..) loop
3. do{ …. } while(..);

Syntax:

for(**initialization** ; condition ; increment / decrement)
{ …… }

can be used, when number of iterations are known.
Pre-condition loop

initialization:

- This statement will run only for the first time when loop will start.
- More than one initialization is allowed separated by comma(,)
- Generally we initialize loop counter here. But it can be used for other variables.
- It is not compulsory to write initialization

Syntax:

for(initialization ; **condition** ; increment / decrement)
{ …… }

Condition:

- This has to be an expression which returns true or false
- Only one condition expression has to be there
- The condition will be checked every time and loop statements will keep on executing if condition is true, if the condition expression is false, the loop terminates.
- It is not compulsory to write condition. In case condition is not written, programmer should use break statement to terminate loop, otherwise it will cause infinite loop.

Syntax:

for(initialization ; condition ; **increment / decrement**)
{ …… }

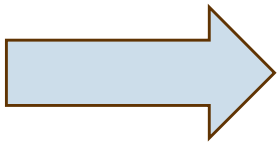
Increment / decrement:

- This will execute every time after first iteration before checking the condition.
- More than one statements are allowed separated by comma (,)
- It is not compulsory to write.

Sample code to understand for loop

```
#include<iostream>
using namespace std;
int main()
{
	int i;
	for(i=1; i<=10; i++)
		cout<<"  "<< i;

	return 0;
}
```



Step - 1. i = 1
Step – 2. check condition i<=10
Step – 3. execute cout i statement
Step – 4. increase value of i (i++)
Step – 5. check condition continue step 3 to 5 until condition becomes false

Syntax:

```
while(condition)
{
    statements;
}
```

Execute loop statement for each time if condition is true.

Pre-condition loop.

Sample code to understand while loop

```
#include<iostream>
using namespace std;
int main()
{
	int i=1;
	while(i<=10)
	{
		cout<<"  "<<i;
		i = i+1;		/*Statement that changes the value of condition*/
	}
	return 0;
}
```

Syntax: do..while loop

```
do {
        …
} while (condition);
```

It will execute at least one time irrespective of condition.

If condition is true it will execute the loop statements otherwise terminate the loop.

Post-condition loop

Sample code to understand do .. while loop

```
#include<iostream>
using namespace std;
int main()
{
	int i=1;
	do
	{
		cout<<"  "<<i;
		i = i+1;
	}while(i<=10);

	return 0;
}
```

Basic assignments on condition and for loops :

1. Write a program in C++ to print Hello 5 times each on different line
2. Write a program in C++ to print all numbers from 1 to 100
3. Write a program in C++ to accept a number and print table of that number
4. Write a program in C++ to accept a number and print factorial of that number
5. Write a program in C++ to accept total terms to print Fibonacci series.
6. Write a program in C++ to print square of all numbers from 1 to 25
7. Write a program in C++ to print all even numbers from 0..200
8. Write a program in C++ to print all odd numbers from 1..200
9. Write a program in C++ to print sum of all numbers from 1 to 100
10. Write a program in C++ to print sum of squares of all numbers from 1 to 25
11. Write a program in C++ to accept 15 numbers from user and print maximum
12. Write a program in C++ to accept 15 numbers from user and print minimum
13. Write a program in C++ to print Fibonacci series of 3 terms.
14. Write a program in C++ to accept a number and print factors of that number
15. Write a program to accept a number and print whether the number is Prime or not.

Basic assignments on condition and while loops :

1. Write a program to accept a number from user and print sum of its digits.
2. Write a program to accept a number and print reverse of that number.
3. Write a program to accept a number and print whether it is Palindrome or not
4. Write a program to accept a number and print whether it is Armstrong or not
5. Write a program to accept a number and print whether it is Perfect number or not.

Basic assignments on switch..case and do..while loops :

1. Write a menu driven program to display menu
    1. Addition
    2. Subtraction
    3. Multiplication
    4. Division
    5. Exit

    Accept choice from user and use switch..case to perform appropriate operation depending on choice user has entered, program should display menu again and accept choice from user until users gives choice to exit from program

2. Write a menu driven program to display hotel menu and calculate total bill.

Nested Loops: Loop with-in loop

```
 for(..;..;..)
{
  for(..;..;..)
  {
       ……
   }
}
```

For each iteration of outer loop, all iterations of inner loop will execute.

We can write any loop structure with in any other loop structure.

Basic assignments on nested loops :

1. Take range of numbers from user and print tables of all numbers within the given range.
2. Print all prime numbers between given range.
3. Menu driven program for small hotel menu with sub menu and print total bill amount, when user enter Exit choice.
4. Take a range of numbers and print all perfect numbers within the given range.